

# Nicholas Ashford

Software Engineer · Distributed Systems, Cloud Infrastructure & Agentic AI

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I'm a software engineer and systems programmer with 18+ years of experience, currently CTO of Education Intelligence, the research business behind Teacher Tapp and School Surveys, where the platform is built in TypeScript, Go and Rust. I own its infrastructure end to end: fault-tolerant, idempotent microservices over message brokers, running on Kubernetes across AWS, Google Cloud and Azure. Over the past year I've also rebuilt the company to be AI-native: agents now implement our tickets, sit in our planning calls, and connect to every system in the business. I'm always happy to be handed a challenge or something broken, and to take it apart to find out exactly why.

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## Highlights

- **Systems programming:** fault-tolerant, idempotent microservices in Go and Rust that absorb extremely spiky traffic; real-time Navier-Stokes fluid simulation on the PlayStation 3 GPU.
  - **Infrastructure:** owned cloud infrastructure across AWS, Google Cloud and Azure, built with Terraform and Kubernetes.
  - **Custom integrations:** energy-provider integrations at Switcher.ie and uSwitch, on-prem data extraction inside schools, MCP proxies for closed SaaS products.
  - **Agentic delivery:** built a multi-agent software factory pipeline, with productivity gains that have let the business absorb natural attrition without replacement hires.
  - **Voice agent:** joins project management calls and raises contextualised tickets from the conversation.
  - **MCP infrastructure:** AI in the hands of everyone in the business, with custom MCP connectors and proxies for the systems that lacked them.
  - **Early CI:** built an automated pull-request-to-production pipeline for 100 engineers at uSwitch, before mainstream CI tooling existed.
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## Experience

### Lead Engineer & CTO

#### Education Intelligence (Teacher Tapp & School Surveys)

January 2021 – Present

*Education research business running the UK's largest daily teacher survey panel: thousands of teachers respond every day, and the resulting data is used by policymakers, academics and the national press.*

- Rebuilt the Teacher Tapp app from an MVP into TypeScript and React Native, establishing test-driven development and agile delivery as standard practice.
- Took the School Surveys MVP to a production-grade platform in TypeScript and Rust.
- Architected both platforms as fault-tolerant, idempotent microservices communicating over message brokers, with queue consumers written in Go.
- Now responsible for two Rust backend platforms and nine TypeScript applications across the two businesses.
- Designed and built a multi-agent pipeline: a workflow of agents implements tickets from specifications and acceptance criteria, raises pull requests, and ships only after automated tests and human review.

- Built a voice agent that joins project management calls, listens to the discussion and raises contextualised tickets, replacing the Kanban board.
  - Rolled AI out to every member of the business and connected its systems to Claude via MCP, including a proxy service that exposes MCP connectors for SaaS products that lack them.
  - Built and led a team of engineers, mentoring each of them through to more senior roles.
  - Responsible for security: protecting sensitive survey and school data, hardening infrastructure, and handling security testing and remediation.
  - Established the end-to-end product process (specification, design through a retained agency, MVP build, release and analytics) that takes products from concept to launch.
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## Senior Software Engineer

### Switcher.ie

February 2019 – January 2021 · Dublin

- Part of a small team of senior engineers building the service-oriented platform behind Ireland's leading comparison site, handling lead capture from the website.
  - Designed and built the framework for distributing leads to the utility providers.
  - Owned the infrastructure: preview, staging and production environments and their monitoring, built with Terraform and Kubernetes on AWS.
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## Lead Software Engineer

### Assembly Edu

January 2016 – March 2019 · London

- Helped establish the company: structured as a 50/50 partnership between NEON's investment arm and an education provider, with the platform built from the ground up.
  - Designed and built the MIS Connector, a .NET ETL system running on servers inside schools, extracting sensitive management information data and syncing it to the cloud.
  - Wrote the cloud processing pipeline that turned raw school data into a queryable database, consumed by third-party education providers for analysis.
  - Owned the infrastructure (AWS) and a substantial share of the Ruby platform.
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## Senior Developer

### NEON Adventures

January 2014 – January 2017 · London & New York

- Consulting and tech leadership on client projects for a venture studio, spanning charities and consumer brands.
  - Built interactive 3D visualisations for a professional cycling team's website, at a time when 3D in the browser was still rare.
  - Built backends for charity clients including kindness.org and Charity: Water, plus a client-management system for a further charity.
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## Senior Graphics Programmer

### Sony Computer Entertainment Europe

February 2012 – January 2014 · London

- Graphics programmer on Harry Potter titles for PlayStation 3 (C++/GPU), on the team responsible for special effects and integrating the art-tool pipeline.
  - Built a real-time fluid simulation for the potion cauldron, implementing the Navier-Stokes equations in shaders on the PS3 GPU.
  - Implemented augmented-reality features using the PlayStation Eye, projecting a handheld device into in-game 3D environments.
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## Senior Developer

### Forward Internet Group (uSwitch)

July 2008 – February 2012 · London

- Full-stack development on uSwitch.com's gas and electricity comparison products: lead capture, processing and integrations with energy providers.
  - Built the company's internal deployment system: pull requests from up to 100 engineers triggered an automated pipeline through to production.
  - Founded and ran an iOS games studio within Forward's incubator: hired the team, then led development, release and marketing of a C++ iOS game over the course of a year.
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## Skills & Technologies

- **Cloud & infrastructure:** AWS, Google Cloud Platform, Azure, Terraform, Kubernetes, event-driven architecture, message brokers, CI/CD pipelines, preview/staging/production environment provisioning, monitoring
  - **AI & agentic engineering:** LLM orchestration and prompt engineering, agentic development workflows, multi-agent pipelines, MCP connector and proxy development, generative AI, machine learning, PyTorch
  - **Languages:** TypeScript, Go, Rust, C++, Python, JavaScript, C#, Ruby, Java, SQL
  - **Graphics & simulation:** GPU and shader programming, real-time fluid simulation, 3D mathematics, augmented reality
  - **Data & integrations:** ETL pipelines, on-premise data extraction, third-party API integrations, Microsoft SQL Server, large-scale data processing
  - **Web & mobile:** React, Node.js, .NET, iOS, Android
  - **Leadership & practice:** team building and mentoring, agile delivery, test-driven development, service-oriented architecture
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## Location

British citizen currently based in Dubai, with EU work rights through spouse.